

Mayank Rath

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Education

- M.Sc. Economics** Jul 2025 - Present
Indian Institute of Technology Kanpur (IITK), *CGPA: 7.86/10*
Relevant coursework: Econometrics, Economic Analysis, Machine Learning, Game Theory, Mathematical Finance
- B.Sc. (Hons.) Mathematics** Oct 2021 - Jun 2024
University of Delhi, *CGPA: 8.162/10*
Relevant coursework: Microeconomics, Macroeconomics, Real Analysis

Professional Experience

- Research Assistant** Dec 2025 - Present
Department of Economic Sciences, IIT Kanpur Kanpur, India
- Developed optimization-driven Stackelberg game models across 2 research initiatives to evaluate strategic interactions, supporting evidence-based policy analysis and equilibrium decision making.
 - Benchmarked 4 optimization algorithms for equilibrium computation, improving model interpretability, computational efficiency, and robustness across complex economic systems.
 - Synthesized technical findings into executive-ready presentations and research documentation, enabling faster stakeholder reviews and supporting timely academic decision making.

- Research Analysis Intern** Sep 2024 - Jan 2025
Department of Electronics, ANDC, University of Delhi Delhi, India
- Identified fragmented reporting processes across a large-scale STEM initiative by analyzing 1,100+ educational records using Python and SQL; automated reporting pipelines and Power BI dashboards, reducing reporting turnaround time by 30%.
 - Evaluated 15+ academic and behavioral variables to uncover performance gaps across participant cohorts, generating actionable recommendations adopted by 3 academic departments.
 - Standardized KPI reporting workflows for 20+ stakeholders, eliminating manual data consolidation and improving consistency in program evaluation and decision making.

Leadership and Activities

- Student Ambassador | Wolfram Inc.** Jun 2024 - Mar 2026
- Expanded adoption of computational tools across 3+ academic departments by delivering training sessions in Python and Mathematica, improving analytical capabilities for STEM students.
 - Designed hands-on workshops covering statistical modeling, visualization, and computational problem solving, strengthening data analytics proficiency among student participants.

- President, SPIE Student Chapter | University of Delhi** Jun 2023 - Jun 2024
- Scaled the student chapter from inception to 150+ active members by restructuring operations across 4 functional teams and implementing structured leadership workflows.
 - Directed 21 academic, industry, and outreach events engaging 2,000+ participants, strengthening institutional visibility and industry collaboration.
 - Led cross-functional coordination of faculty, student, and industry stakeholders, contributing to the chapter receiving the SPIE International Presidential Award Runner-up (2023) among 340+ global chapters.

Projects

- AI Execution Gap: Why Enterprise AI Projects Underperform** May 2026 - Jun 2026
- Analyzed 300 enterprise AI deployments across 8 industries and 5 regions to identify organizational factors driving project success and failure.
 - Quantified the impact of 19 governance and operational variables, identifying poor data quality (40 projects), insufficient budgets (37 projects), and siloed organizational structures (31 projects) as the strongest failure drivers.
 - Developed an executive dashboard summarizing adoption trends, organizational readiness, and investment performance, highlighting failure rates of 73.08% in 2022 before improving to 62.50% in 2024.
 - Designed a 4-step enterprise AI adoption framework translating analytical findings into strategic recommendations for governance, talent development, and change management.

- Retail Pricing Elasticity and Promotion Strategy** Feb 2026 - Mar 2026
- Estimated price elasticity across 10+ product categories using log-log regression models to identify demand responsiveness and pricing inefficiencies.
 - Engineered 15+ pricing and behavioral features, including promotional frequency, discount depth, and competitor pricing, to improve elasticity estimation accuracy.

- Segmented product categories into 4 pricing strategies through scenario simulations, identifying over-discounting of 12–18% across high-volume categories.
 - Built Tableau dashboards translating elasticity estimates into pricing recommendations, simulating 15% higher contribution margins compared with blanket discount strategies.
- UPI Merchant Profitability & MDR Policy Simulation for India’s Digital Payments Ecosystem** Nov 2025 - Dec 2025
- Evaluated the financial sustainability of India’s zero-MDR UPI ecosystem by analyzing 5+ years of NPCI and RBI payment data, assessing profitability across merchants, banks, and payment service providers.
 - Developed a scenario-based pricing model simulating multiple MDR structures (0%, 0.1%, 0.25%, and tiered pricing), quantifying trade-offs between digital payment adoption, stakeholder profitability, and transaction growth.
 - Built an executive decision-support dashboard integrating transaction volumes, operating costs, elasticity estimates, and profitability metrics to recommend data-driven MDR strategies under different regulatory scenarios.

- Customer Segmentation – SME Client Strategy** Aug 2025 - Sep 2025
- Analyzed 2,000+ SME customer records across 15 financial and behavioral variables to identify revenue leakage caused by low-precision targeting.
 - Developed a K-Means clustering pipeline generating 5 actionable customer segments, improving campaign targeting efficiency by 25%.
 - Delivered segment-specific recommendations and interactive dashboards that increased campaign conversion rates by 22% through data-driven marketing strategies.

- Skills**
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- **Business & Strategy:** Business Analysis, Competitive Analysis, Market Research, Financial Analysis, Strategic Planning, Scenario Analysis, Root Cause Analysis, Stakeholder Management, Cross-functional Collaboration
 - **Data Analytics & BI:** Python, SQL, Power BI, Tableau, Excel (Advanced), KPI Reporting, Dashboarding, Data Visualization
 - **Statists & Economics:** Econometrics, Time Series Analysis, Regression, Bayesian Inference, Forecasting, Hypothesis Testing, Risk Analytics, Operations Research
 - **Machine Learning:** Predictive Analytics, Customer Segmentation, Clustering, Classification, Feature Engineering, Predictive Modeling, Optimization

Certifications

Business Analytics Specialization (Wharton) | Data Analytics Essentials (Cisco) | Machine Learning (Stanford)

- Publications**
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1. Innovative STEM pedagogies for teaching trigonometry and mensuration concepts, *Physics Education (IOP)*, 2025.
 2. Analysing the success of mega optics outreach program using mathematical methods, *MGM Publishing*, 2024.